

Movement

Snakes have evolved several ways of moving around (see right) to compensate for not having legs. The method used involves different uses of their ribs and the muscles attached to them, but also depends on the snake's weight, the speed at which it needs to move, and the type of surface. Most snakes can use different types of locomotion as the need arises. As well as the methods shown here, some vipers use a technique called sidewinding when crossing loose sand (see p.406).

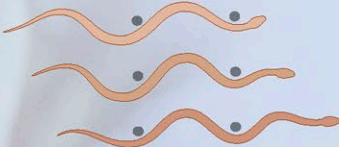
LINEAR PROGRESSION

Heavy snakes and snakes that are moving slowly or stalking prey sometimes progress in a straight line, hooking the trailing edges of the ventral scales over surface irregularities. This occurs in a wavelike sequence, as the snake pulls itself forward.



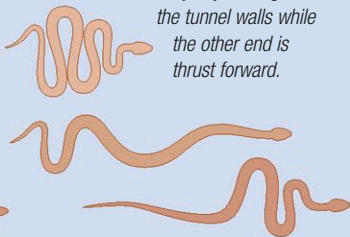
LATERAL UNDULATION

Using each point of its body in turn to push against irregularities in the surface, the snake wiggles from side to side. This method is also used for swimming and sometimes climbing.



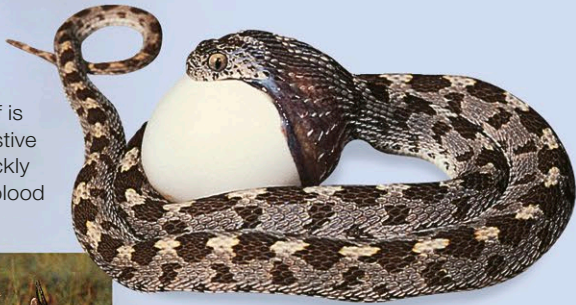
CONCERTINA MOVEMENT

When moving through their tunnels, burrowing snakes expand one end of the body to jam it against the tunnel walls while the other end is thrust forward.



Hunting and feeding

Some snakes are specialized feeders, while others eat a wide range of prey, including other snakes. Small animals or ones that cannot fight back are simply grasped and then swallowed; animals that are larger or better able to defend themselves must be subdued with venom or by constriction before being eaten. A large meal may take a long time to digest, starting in the mouth with powerful salivary juices. Venom itself is a cocktail of modified digestive juices that usually acts quickly on the nervous system or blood tissue of the prey.



SWALLOWING

Flexible jaws and highly elastic skin allow snakes, like this common egg-eater, to eat prey that is larger than their own head. A large meal can take several hours to swallow.

CONSTRICTION

A constrictor kills its prey by suffocation. Each time its victim breathes out, this rock python tightens its coils until breathing stops.



Reproduction

Most snakes lay eggs, but a significant number give birth to live young. Species from temperate regions usually mate in spring, soon after emerging from hibernation, and produce eggs or young in summer. Tropical species may breed in response to rainfall and sometimes have a long breeding season, laying several clutches of eggs each year. Males and females find each other using chemical scent trails, and courtship is rarely elaborate. Snakes show little parental care, although pythons and a few other species coil around their incubating eggs.



EGGS AND LIVE YOUNG

Young snakes, like this monocled cobra, break out of their shell using a sharp, temporary egg tooth. The snake is often coiled tightly inside its shell, and may be up to 7 times longer than the egg.

CLIMBING

To enable them to move easily through their habitat, tree snakes have a thinner, more lightweight body than ground-dwelling or burrowing snakes. They have long tails that can be used to grasp branches, and strong vertebrae that help them to cross wide gaps. The pit viper shown here (Schultze's pit viper) is a nocturnal hunter that uses its large eyes and its heat pits to locate prey.



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SWALLOWING HEAD FIRST

The gecko is swallowed head first because the limbs fold more easily in this direction.



6

EXTENDING THE JAWS

The snake opens its highly flexible jaws and uses its rear fangs to pull the gecko into its mouth.



7

MUSCULAR ACTION

When most of the prey is in the throat, wavelike muscle contractions force it into the stomach.



8

DIGESTION

Having completely swallowed its prey, the snake searches for a quiet place to digest its food.